

# SQL Data Analysis Summary

Project 3: SQL Data Analysis, DecodeLabs Data Analytics Internship

## Objective

Use SQL queries to extract insights from the cleaned e-commerce order dataset (1,200 records). Demonstrate SQL fundamentals: SELECT statements, filtering (WHERE), sorting (ORDER BY), grouping (GROUP BY), and aggregation (COUNT, SUM, AVG), plus filtering on aggregated results (HAVING) and percentage-contribution analysis.

## Environment

The cleaned dataset was exported to CSV and imported into a SQLite database (table: dataset\_for\_data\_analytics, 1,200 rows) using DB Browser for SQLite.

### Query 1: Basic SELECT

```
SELECT * FROM dataset_for_data_analytics LIMIT 10;
```

### Query 2: WHERE (filter rows)

```
SELECT OrderID, Product, Quantity, TotalPrice
FROM dataset_for_data_analytics
WHERE OrderStatus = 'Delivered';
```

### Query 3: ORDER BY (sort results)

```
SELECT OrderID, Product, TotalPrice
FROM dataset_for_data_analytics
ORDER BY TotalPrice DESC
LIMIT 10;
```

### Query 4: GROUP BY + COUNT (orders per product)

```
SELECT Product, COUNT(*) AS OrderCount
FROM dataset_for_data_analytics
GROUP BY Product
ORDER BY OrderCount DESC;
```

| Product | Order Count |
|---------|-------------|
| Printer | 181         |
| Tablet  | 179         |
| Chair   | 178         |
| Laptop  | 173         |
| Desk    | 170         |
| Monitor | 163         |
| Phone   | 156         |

### Query 5: GROUP BY + SUM (revenue per product)

```
SELECT Product, SUM(TotalPrice) AS TotalRevenue
FROM dataset_for_data_analytics
GROUP BY Product
ORDER BY TotalRevenue DESC;
```

| Product | Total Revenue (\$) |
|---------|--------------------|
| Chair   | 195,620.11         |
| Printer | 195,612.61         |
| Laptop  | 192,126.56         |
| Tablet  | 186,568.95         |
| Monitor | 175,651.41         |
| Desk    | 167,459.93         |
| Phone   | 151,722.39         |

**Query 6: GROUP BY + AVG (average order value by payment method)**

```
SELECT PaymentMethod, AVG(TotalPrice) AS AvgOrderValue
FROM dataset_for_data_analytics
GROUP BY PaymentMethod
ORDER BY AvgOrderValue DESC;
```

| Payment Method | Avg Order Value (\$) |
|----------------|----------------------|
| Credit Card    | 1,127.55             |
| Gift Card      | 1,070.97             |
| Cash           | 1,056.04             |
| Online         | 1,017.22             |
| Debit Card     | 1,001.56             |

**Query 7: HAVING (filter on aggregated results, bonus)**

```
SELECT Product, COUNT(*) AS OrderCount, SUM(TotalPrice) AS TotalRevenue
FROM dataset_for_data_analytics
GROUP BY Product
HAVING SUM(TotalPrice) > 190000
ORDER BY TotalRevenue DESC;
```

| Product | Order Count | Total Revenue (\$) |
|---------|-------------|--------------------|
| Chair   | 178         | 195,620.11         |
| Printer | 181         | 195,612.61         |
| Laptop  | 173         | 192,126.56         |

**Query 8: Percentage contribution to total revenue (bonus)**

```
SELECT Product,
       SUM(TotalPrice) AS TotalRevenue,
       ROUND(SUM(TotalPrice) * 100.0 /
             (SELECT SUM(TotalPrice) FROM dataset_for_data_analytics), 2) AS PctOfTotalRevenue
FROM dataset_for_data_analytics
GROUP BY Product
ORDER BY TotalRevenue DESC;
```

| Product | Total Revenue (\$) | % of Total Revenue |
|---------|--------------------|--------------------|
| Chair   | 195,620.11         | 15.47%             |
| Printer | 195,612.61         | 15.47%             |

|         |            |        |
|---------|------------|--------|
| Laptop  | 192,126.56 | 15.19% |
| Tablet  | 186,568.95 | 14.75% |
| Monitor | 175,651.41 | 13.89% |
| Desk    | 167,459.93 | 13.24% |
| Phone   | 151,722.39 | 12.00% |

## Key Findings

- Order volume is fairly evenly spread across all 7 products (156–181 orders each), with Printer leading by count (181) and Phone trailing (156).
- Revenue ranking shifts meaningfully from volume ranking: Chair overtakes both Printer and Tablet to become the #1 revenue-generating product (\$195,620.11) despite ranking only 3rd in order count. This indicates Chair carries a higher average order value than its purchase frequency alone would suggest.
- Phone is the consistent underperformer across every metric measured, lowest order count (156), lowest total revenue (\$151,722.39), and lowest share of total revenue (12.00%).
- Revenue contribution is remarkably balanced, no single product dominates. The top 3 products (Chair, Printer, Laptop) each contribute between 15.19% and 15.47% of total revenue, a near-even three-way split at the top of the range.
- Payment method has a modest but real effect on order value, Credit Card orders average \$1,127.55, roughly 12.6% higher than Debit Card orders (\$1,001.56), the lowest-spending method. The spread across all five payment methods is narrow (\$1,001–\$1,128), suggesting payment method is a minor factor in order size, not a major driver.
- The HAVING clause correctly isolated the three products exceeding \$190,000 in total revenue (Chair, Printer, Laptop), demonstrating filtering on an aggregated result, which WHERE cannot do since WHERE operates on raw rows before grouping occurs.

## Business Relevance

Chair should be treated as a priority product despite its mid-tier order volume, since it drives disproportionately high revenue per order, a strong candidate for upsell or bundling strategies. Phone consistently underperforms across all measures and may warrant review (pricing, marketing, or placement). Payment method is not a strong lever for increasing order value and should be deprioritized relative to product-level strategies.